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Are We Really Transitioning?

Getting to Know Renewable Energy Certificates: A Solution or a Shortcut?

By Fisaniya 'Sanya' Rahmanisa



After nearly a year of immersing myself in sustainability reports, one claim has increasingly caught my attention: 100% renewable electricity. The phrase sounds convincing. However, I began to wonder if a factory still receives electricity through the same grid as other factories, what actually changes when a company buys renewable energy certificates?

That question resurfaced when I attended a forum on the theme of *Renewable Energy Zone*. There, I found myself among members of the media, delegates, investors, organizers, and VVIP guests. At first, I came without any particular expectations. I simply wanted to take in as much of the discussion as I could.

The discussion on renewable energy and carbon emissions accounting, particularly Scope 2 and Scope 3, quickly caught my attention. When green solutions began to be discussed, the greenwashing alarm in my head briefly went off. However, I held it back and continued to listen to the rest of the session with my suspicions on the matter.. I then decided to look into the issue further. One instrument that made me curious was the Renewable Energy Certificate (REC). From the outside, a REC looks simple- a certificate that helps companies claim the use of renewable electricity. However, when viewed from the perspectives of marketing, finance, and sustainability, the story turns out to be far less simple.

Marketing: When a Certificate Becomes a Claim

RECs can be an effective communication tool. According to the OECD, one certificate represents the renewable energy attributes of electricity generated in a certain amount, typically one megawatt-hour (MWh). Companies can purchase and use these certificates to support claims of renewable electricity use in their emissions reporting. For companies, such instruments offer a way to claim renewable energy attributes without having to build their own power plants.However, the issue lies not only in the certificate itself, but also in how companies communicate their purchase.

The statements "we purchase RECs to cover our electricity consumption" and "our factory runs on renewable electricity" can create different impressions. The first statement explains the instrument the company uses. The second may lead readers to imagine that all of the electricity physically flowing into the factory comes from renewable power plants.

In fact, purchasing RECs does not in itself change the source of the electricity a facility physically receives. A REC relates to renewable energy attributes that are traded and claimed by the certificate holder; it is not a guarantee that the electrons reaching the factory come from a particular power plant. Situmorang (2025) also notes that green certificates are not the kind of certificate granted through review and assessment by an authorized body because the system is market-based.

This, in my view, is where the issue of *greenwashing* needs to be examined. RECs are not automatically a form of *greenwashing*. However, the claims built on them can become misleading if they do not proportionately explain what the company is actually doing.

In other words, the certificate may be legitimate but the claim that accompanies it still needs to be verified.

Finance: How Strong Is the Incentive to Transition?

From a financial perspective, RECs offer something attractive: a relatively low cost of supporting renewable energy claims.

Situmorang (2025) stated that the price of one REC unit, equivalent to 1,000 kWh, is Rp35,000. This means the certificate costs Rp35 per kWh. As a simple illustration, a factory that uses 1 GWh of electricity per year would need only about Rp35 million to buy RECs covering its entire consumption. Compared with the investment needed to build its own power plant, RECs can be a more affordable option for companies that want to meet their electricity emissions reporting needs.

The market is also growing. Situmorang reported that PLN's green certificate sales reached 13.68 TWh

in the first half of 2025. For PLN, this market opens up revenue opportunities from renewable energy attributes. For buyers, RECs offer an instrument to support renewable energy claims without having to develop their own power plants.However, growing certificate sales do not immediately answer the question of their impact on the energy transition.

Affordable prices cut both ways. On one hand, they can lower the barriers for companies that want to start using renewable energy instruments. On the other hand, they raise the question of how strong an economic incentive they create to build new renewable energy capacity.

In the OECD discussion, corporate buyers emphasize the need for REC products that are affordable, internationally recognized, and able to demonstrate a credible contribution to renewable energy development. Developers, for their part, considered demand certainty, long-term offtake, price transparency, as well as regulatory certainty essential for REC revenue to be financeable (*bankable*) and to support investment.

Both perspectives remind us of two interests that do not always align: affordability for buyers and revenue certainty for developers.

I myself have not yet found sufficient grounds to conclude that low REC prices necessarily weaken the drive for renewable energy investment. Price is not the only factor that determines investment decisions. Still, the question remains relevant: does purchasing RECs merely help companies manage their electricity emissions claims, or does it also provide a meaningful push toward adding clean energy supply?

To answer this, we need to look beyond certificate prices and sales volumes. We need to know whether that demand support renewable energy capacity that would not otherwise have been built.

Sustainability: What Actually Changes?

The question of impact becomes more important when RECs are placed in the context of Indonesia's energy transition.

A *policy brief* from the ASEAN Centre for Energy on the REC market in Indonesia highlights the need for a clearer national governance framework. The study notes the absence of binding national regulations for renewable energy and recommends a legal framework for RECs with a clear national authority. A renewable energy legal guide from the law firm ABNR likewise does not mention any specific legal basis for RECs.

The ASEAN Centre for Energy *brief* also notes that I-REC issuance in Indonesia fell from 1.76 million MWh in 2023 to 1.44 million MWh in 2024. Meanwhile, the number of certificates used, or redeemed, rose from 1.31 million to 1.43 million MWh.

These figures show that certificate issuance and use do not always move in the same direction. However, the figures alone are not enough to explain the causes of these changes or to measure the impact of RECs

on additional renewable energy capacity.

Governance issues also arise in discussions of energy-intensive industries. The same *brief* highlights the existence of 116 nickel smelters in 2023, a sector that requires a stable electricity supply and faces challenges in reducing its dependence on fossil-based energy. In the context of international trade, the emissions intensity of production is also becoming increasingly relevant as climate policies such as the European Union's *Carbon Border Adjustment Mechanism* (CBAM) develop.

RECs can help companies document the renewable energy attributes they claim in their reporting or for certain supply chain requirements. ABNR states that RECs currently serve as proof for exporting companies that their production uses renewable energy. However, owning certificates does not automatically prove that the production process at a facility has physically switched to renewable electricity, let alone that the overall emissions of its production have declined.

This distinction matters. Administrative proof of renewable energy attributes, a change in the source of the electricity used, and actual emissions reductions are three related things, but they are not always identical.

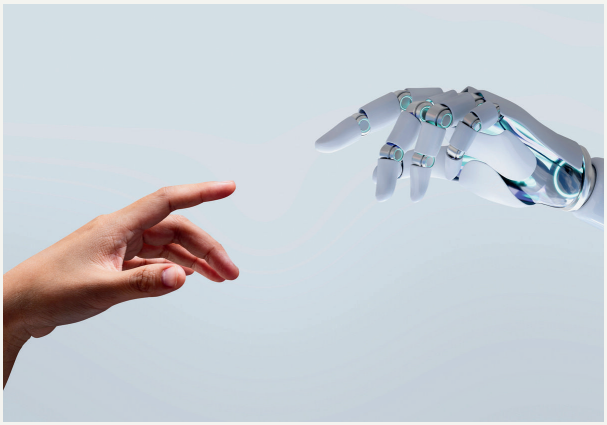
A similar picture emerged in an OECD discussion with the Ministry of Energy and Mineral Resources in Jakarta on May 26, 2026. The summary of that discussion describes Indonesia's REC market as still voluntary and fragmented, with various transaction channels, including PLN's scheme, the ICDX exchange, and direct purchases, and as lacking a





unifying national legal framework. The discussion also highlighted the limited options for purchasing electricity through a *corporate power purchase agreement* (CPPA), which means companies may have to rely on RECs purchased separately from their electricity supply, or *unbundled* RECs.

One of the issues discussed was the risk of *double counting*: the attributes or environmental benefits of the same energy source must not be claimed by more than one party. This means that the credibility of RECs depends not only on the certificates a company holds, but also on market governance, the traceability of energy attributes, and clarity about which claims are permitted.



Can Technology Close the Gap?

Some researchers have proposed *blockchain* as a way to improve the REC system in Indonesia. Haryani et al., writing in the journal Heliyon, describe PLN's current system as relying on a centralized database and consider its transparency and *real-time* data access to be limited. They interviewed four PLN staff members and built a prototype as a *proof of concept*. An IEEE paper written by PLN staff (Effendi et al.) is conceptual in nature and concludes that a *blockchain*-based REC system can improve transparency, traceability, and stakeholder trust.

In my view, this answers the question "Is this certificate genuine and not counted twice?" but does not necessarily answer the question "Has any additional clean energy been created?" Haryani et al. themselves note that the costs and benefits of this solution still require further research.

Conclusion

The OECD summary states that a credible REC framework needs to align with international standards, such as the *GHG Protocol Scope 2 Guidance* and the RE100 technical criteria. I have not yet looked into how these standards are evolving, so I will not discuss them further here. Having looked at RECs from the perspectives of *marketing*, *finance*, and *sustainability*, I do not see them as entirely good or entirely bad.

RECs can help companies claim renewable energy attributes, support electricity emissions reporting, and provide access to a market instrument that is relatively easy to use. However, the certificates are not automatic proof that a company has fully decarbonized. Moreover, purchasing RECs does not in itself prove that any additional renewable energy capacity has been created.

For that reason, I think RECs are better seen as a bridge, not the final destination.

When we come across claims of renewable electricity use in a company's sustainability report or promotional materials, there are three simple questions we can ask.



From a *marketing* perspective:

Does the company transparently explain that its claim is supported by REC purchases, or does it instead give the impression that all the electricity it uses physically comes from renewable power plants?

From a *finance* perspective:

Are RECs part of a broader energy transition strategy, and how does their role compare with direct investment in energy efficiency, electrification, or renewable energy development?

From a *sustainability* perspective:

Does the purchase help drive additional clean energy supply that would not otherwise have been available, and how does the company prove its environmental contribution?

These three questions are not intended to reject RECs. On the contrary, they help us assess this instrument more clearly: not only by the claims it generates, but also by what it can actually prove.

Ultimately, the energy transition requires more than the ability to calculate and communicate emissions. It requires traceable change, consistent investment, and claims that match actions.

RECs can indeed be a bridge. What we need to examine is whether companies are actually crossing it or merely posing for photos in the middle of it.